



PCET's

Pimpri Chinchwad College of Engineering

Department of Information Technology

B. Tech Project



Problem Statement : Existing electronic and traditional voting systems often prioritize security over usability, making them difficult for elderly, disabled, and low-literacy voters to use. They also face challenges such as lack of transparency, risk of tampering, limited verifiability, and potential privacy concerns due to linkage of voter identity with votes. Hence, there is a need for a secure, transparent, and accessibility-first voting system that ensures voter privacy, prevents fraud, and enables inclusive participation.

SDG Mapping : SDG 9: Industry , Innovation and Infrastructure
SDG 16 : Peace , Justice and Strong Institution

Abstract : VoteGuard Pro is an accessibility-first electronic voting system designed to ensure secure, transparent, and inclusive elections. It uses session-based identities, cryptographic security, and tamper-proof audit logs to maintain voter privacy and system integrity. The system integrates decentralized principles to enhance trust while providing a user-friendly interface for all voters. This approach improves participation, verifiability, and overall confidence in the electoral process.

Aim

To design and develop a secure, accessible, and verifiable electronic voting system that ensures voter privacy, transparency, and inclusive participation for all users.

Objectives

- To develop a user-friendly and accessibility-first voting interface for diverse users.
- To ensure secure voter authentication without linking identity to votes.
- To implement tamper-proof vote recording using cryptographic techniques.
- To provide end-to-end verifiability and auditability of the voting process.

Methodology/Approach

- **System Design:** Develop a modular architecture separating authentication, voting, and vote storage.
- **Voter Authentication:** Verify users using ID/biometric methods while ensuring no linkage to the vote.
- **Ballot Interface:** Provide an accessible, user-friendly voting interface with error prevention.
- **Secure Voting:** Record votes using encryption and hash-based techniques for tamper resistance.
- **Audit & Verification:** Maintain a secure, append-only audit log for transparency and verification.
- **Result Processing:** Ensure accurate and reliable vote counting with integrity checks.

Applications

- **Government Elections** – Secure and transparent voting for local, state, and national elections.
- **Institutional Voting** – Use in colleges, universities, and organizations for elections.
- **Corporate Decision Making** – Board voting and shareholder decision systems.
- **Remote & Online Voting** – Enables voting for citizens in remote or overseas locations.
- **Accessible Voting Systems** – Supports inclusive voting for elderly and differently-abled users.

Results/Outcomes

- Ensures **secure and private voting** using encryption and session-based identities.
- Provides **accessible and user-friendly interface** for all voters.
- Enables **tamper-proof and verifiable voting** through audit logs and blockchain concepts.
- Prevents duplicate voting with **reliable authentication mechanisms**.
- Improves **transparency, trust, and overall efficiency** of the electoral process.



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